

Analyzing Demand for AI Skills

Data Analysis Case Study

Analyst: Vansh Dhiman

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Analyst Role	Junior Data Analyst
Dataset	Global AI & Data Science Job Market
Framework	Ask → Prepare → Process → Analyze → Share → Act
Goal	To identify the most in-demand AI skills and career opportunities in the European job market.

Phase 1 — Ask

Business Task

Career advisors and educational institutions need evidence-based insights into AI hiring trends to help students develop relevant skills and improve employability.

Guiding Questions

- Which AI skills are most demanded?
- Which countries and cities have the most AI jobs?
- What skills are required for entry-level positions?
- What skills are associated with higher salaries?
- Which AI roles are growing fastest?

Key Stakeholders

Name	Role	Interest
University Career Centers	Support students with career planning and employment opportunities	Improve graduate employability; understand which AI skills, job roles, and locations are most in demand to provide better career guidance.

Name	Role	Interest
Students	Preparing for AI-related careers	Develop relevant skills and secure internships/jobs; identify valuable skills, certifications, and career paths to increase employability.
Job Seekers	Looking for employment opportunities in AI and data-related fields	Find suitable jobs and improve competitiveness; learn which skills employers seek, where opportunities are, and what qualifications are required.
HR Departments	Hiring and workforce planning	Attract qualified AI talent; understand labor market trends, skill availability, and hiring demand to improve recruitment strategies.
Recruitment Agencies	Connecting employers with qualified candidates	Match talent with market demand; identify high-demand roles, skill shortages, and hiring hotspots to better serve clients and candidates.

Phase 2 — Prepare

Data Source

Global AI & Data Science Job Market (2020-2026) ([Dataset link](#))

ROCCC Evaluation

Criteria	Assessment	Notes
Reliable	The dataset is publicly available through Kaggle and contains structured job market information.	The dataset is almost ready for analysis.
Original	The dataset creator aggregated information from job market sources.	The dataset is uploaded by one of kaggle users.
Comprehensive	The dataset includes multiple variables needed for skill-demand and salary analysis.	The data is about the job trends and skills which are in demand..
Current	[The dataset covers recent AI job market trends.(2020-2026)	The zip file has 5 files .
Cited	The original source and dataset documentation will be referenced in the final report.	Dataset Licence

Data Inventory

Table / File	Rows	Columns	Description
<i>Ai_jobs</i>	50000	14	<i>The table contains the information about Job Posting</i>
<i>Skills_demand</i>	224605	4	<i>The table tells us about which skill jobs are needed.</i>
<i>Country in Trends</i>	42	6	<i>The table tells us about job demands by countries.</i>
<i>Data Dictionary</i>	14	3	<i>This table contains historical metadata for global job advertisements</i>
<i>Job tittle mapping</i>	6	3	<i>This table acts as a bridge to clean up that noise and group similar roles together for high-level reporting.</i>

Phase 3 — Process

Clean and transform raw data into a reliable, analysis-ready dataset.

Tools Selected

Tool	Purpose
<i>Google Coolab Notebook</i>	<i>A notebook is best because we can do multiple tasks together.</i>
<i>[Tool]</i>	<i>[Why this tool for this specific task]</i>
<i>[Tool]</i>	<i>[Why this tool for this specific task]</i>

Cleaning Checklist

#	Task	Done?
1	Check for duplicate rows	Done ▾
2	Verify column data types	Done ▾
3	Handle null / missing values	Done ▾
4	Standardize text formats (names, categories)	Done ▾
5	Clean and convert numeric/date fields	Done ▾
6	Remove irrelevant columns	Done ▾
7	Document each transformation step	Done ▾

Data Quality Findings

Table	Column	Issue Found	Severity
<i>Ai_Jobs</i>	14	<i>Nothing (Data was already cleaned)</i>	<i>Low</i>
<i>Skills_In_Demand</i>	4	<i>Nothing (Data was already cleaned)</i>	<i>Low</i>

Table	Column	Issue Found	Severity
Country_Ai_Trends	6	Nothing (Data was already cleaned)	Low
Job_Tittle_Mapping	3	Nothing (Data was already cleaned)	Low

Final Dataset Summary

Table	Rows Before	Rows After	Notes
Ai_Jobs	50000	50000	The table was already fully cleaned .
Skills_In_Demand	224605	224605	The table was already fully cleaned.
Country_In_Trends	42	42	The Table was already fully cleaned
Job_Mapping_Tittle	6	6	The Table was already fully cleaned
Data_Dictionary	14	14	The Table was already fully cleaned

Phase 4 — Analyze

Uncover trends, patterns, and relationships that answer the business questions.

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Key Questions to Answer

#	Business Question	Expected Output
1	Who hires most?	Bar Chart with percentage
2	How is the market growing?	Line Chart (Comparing Germany vs global in the same chart.)
3	What skills are demanded?	Horizontal bar chart (top 20 skills globally)
4	What skill types dominate?	Donart Chart (With skills category)
5	What do beginners need?	Entry-level skills (horizontal bar)
6	What does Germany want?	Germany skills spotlight (horizontal bar)
7	What roles exist?	Top 15 roles (horizontal bar)
8	What does experience pay?	Salary by experience — Global vs Germany (grouped bar)
9	What roles pay most?	Top 10 highest paying roles (horizontal bar)
10	How is work structured?	Remote vs onsite (donut)

Key Findings

#	Finding	Business Implication
1	AI jobs are spread across major countries, with the UK slightly ahead and Germany holding a strong mid-level share (~16.7%).	Competition for AI talent is global, so companies must compete on salary, flexibility, and skills growth — not location.
2	AI hiring is stable over time globally and in Germany, with no major spikes or drops.	The market is steady, so success depends more on long-term skill building than timing job cycles.
3	Core skills like Python, SQL, and cloud (AWS/GCP/Azure) dominate global demand.	These are baseline requirements — without them, candidates are filtered out early.
4	ML, Cloud, and Programming are the main skill categories in AI roles.	AI jobs require full-stack capability: coding, modeling, and deployment.
5	Entry-level roles already demand tools like SQL, TensorFlow, NLP, and cloud basics.	Even junior roles expect job-ready skills, not just theory.
6	Germany's skill demand closely matches global trends.	Skills are transferable — international candidates are well-aligned with the German market.
7	Key AI roles include Data Scientist, ML Engineer, MLOps Engineer, and Data Analyst.	AI careers split into clear tracks — specialization matters early.
8	Salaries rise with experience; Germany is close to global averages.	Experience directly drives income growth in AI careers.
9	High-paying roles include ML Engineer, Applied Scientist, and Data-focused roles.	Applied, business-impact roles tend to pay more than purely research roles.
10	Work models are split between remote, onsite, and hybrid.	AI jobs offer flexibility, enabling cross-country and remote opportunities.

Phase 5 — Share

Communicate insights clearly through visualizations and structured presentation.

Visualization Plan

Chart	Chart Type	What It Shows	Tool
C1	Bar Chart	AI job postings by country — Germany highlighted in gold	Python (matplotlib)
C2	Line Chart	AI hiring growth year-over-year — Global vs Germany	Python (matplotlib)
C3	Horizontal Bar	Top 10 most demanded AI skills globally with % of postings	Python (matplotlib)
C4	Donut Chart	Skill category distribution — ML, Cloud, Programming share	Python (matplotlib)
C5	Horizontal Bar	Top 15 skills required by entry-level AI jobs globally	Python (matplotlib)

Chart	Chart Type	What It Shows	Tool
C6	Horizontal Bar	Top 15 skills demanded in Germany specifically	Python (matplotlib)
C7	Horizontal Bar	Top 15 most common AI job roles globally	Python (matplotlib)
C8	Grouped Bar	Average salary by experience level — Global vs Germany	Python (matplotlib)
C9	Horizontal Bar	Top 10 highest paying AI roles globally	Python (matplotlib)
C10	Donut Chart	Remote vs onsite vs hybrid work split globally	Python (matplotlib)
Poster	Multi-Postel	6-chart summary poster — full story in one view	Python (matplotlib)

Key Findings & Business Implications :

So. No.	Business Questions	Findings	Implication
Q1	Who hires most?	UK leads slightly; Germany holds ~16.7% share	AI talent competition is global — location alone won't decide hiring
Q2	How is the market growing?	Hiring is stable globally and in Germany — no major spikes	Long-term skill building matters more than timing the market
Q3	What skills are demanded?	Python, SQL, and cloud (AWS/GCP/Azure) dominate globally	These are baseline filters — missing them means early rejection
Q4	What skill types dominate?	ML, Cloud, and Programming lead all categories	AI roles require full-stack capability — not just one dimension
Q5	What do beginners need?	Entry-level already requires SQL, TensorFlow, NLP, cloud basics	Junior roles expect job-ready skills — theory alone won't get you hired
Q6	What does Germany want?	Germany's skill demand mirrors global trends closely	Skills are transferable — international candidates align well
Q7	What roles exist?	Data Scientist, ML Engineer, MLOps, Data Analyst dominate	AI careers have clear tracks — early specialization accelerates growth
Q8	What does experience pay?	Salary rises consistently with experience; Germany near global average	Every year of hands-on experience compounds financially
Q9	What roles pay most?	ML Engineer, Applied Scientist, senior data roles pay highest	Applied, business-impact roles outpay pure research positions

So. No.	Business Questions	Findings	Implication
Q10	How is work structured?	Remote, onsite, and hybrid split fairly evenly	AI jobs offer real flexibility — cross-border applications are viable

Phase 6 — Act

Translate insights into concrete, actionable recommendations.

Top 3 Recommendations

#	Recommendation	Supporting Finding
1	Skill-readiness gap is the headline insight — even entry-level AI roles require production tools (SQL, TensorFlow, NLP, cloud), not just theory	F5
2	Geography isn't a real constraint — Germany mirrors global skill demand, so apply cross-border/remote instead of restricting by location	F1,F6,F10
3	Specialize early — applied roles (ML Engineer, Applied Scientist) outpace research-track roles	F7,F9

Known Limitations

- Dataset covers only 6 countries (Australia, Canada, Germany, India, UK, US) — not a true Europe-wide sample; Germany is your only EU data point.
- Salary in USD, no PPP adjustment — doesn't reflect local cost-of-living differences.
- "AI job" labels come from the dataset's own classification — may overlap with general data roles.
- No job-description text — findings are structural (titles, skills, salary), not contextual.

Deliverables Summary

Deliverable	Phase	Done?
Clear statement of the business task	Ask	Done ▾
Description of all data sources used	Prepare	Done ▾
Documentation of cleaning & manipulation	Process	Done ▾
Summary of analysis and key findings	Analyze	Done ▾
Supporting visualizations	Share	Done ▾

Deliverable	Phase	Done?
Top 3 high-level recommendations	Act	Done ▾